Hassan Mahmood

AI Engineer (Agentic Workflows & LLM Systems)

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 5 years of proven experience at Deloitte delivering production Generative AI, machine learning pipelines, and autonomous agent systems. Specialized in architecting multi-agent workflows with LangGraph and MCP, implementing automated evaluation frameworks (DeepEval, Ragas), and deploying reliable LLM pipelines with deterministic guardrails. Strong track record of translating complex enterprise requirements into high-accuracy, production-ready AI solutions.

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, Multi-Agent Systems, MCP (Model Context Protocol), Tool Calling, Prompt Engineering, Fine-Tuning (PEFT/LoRA)

RAG & Retrieval: RAG Architecture, FAISS, Vector Embeddings, Semantic Search, Document Indexing

Machine Learning & Evaluation: PyTorch, Scikit-Learn, DeepEval, Ragas, LLM Guardrails, AI Safety

Software & Deployment: Python, FastAPI, SQL, Pandas, NumPy, Pydantic, Docker, Git, Streamlit, REST APIs

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer – Deloitte — **Jul 2021 -- Present**

- Engineered enterprise AI systems, RAG pipelines, and automated reasoning workflows across 3 major AI initiatives using Python, LangGraph, MCP, and FastAPI.
- Architected 'FinPilot AI', a multi-agent financial analysis platform utilizing Deloitte internal LLMs and LangGraph ReAct; supervisor agent managed state and routing across 4 intent categories with deterministic guardrails.
- Engineered an MCP-based multi-format ingestion workflow using FastMCP, SQLite, and local file drivers to process 3 file formats, reducing ingestion latency by ~80%.
- Developed a RAG and calculation engine combining FAISS-based semantic search with deterministic Python tax logic, achieving a 93% evaluation score on DeepEval benchmarks.
- Architected 'ALAN-GPT', a multi-document RAG search engine across 500+ production ML models, indexing documentation and reducing developer lookup time by 75% with a 91% retrieval score.
- Developed a Generative AI pipeline processing 1,000+ audio call recordings into structured call scripts and summaries, reducing manual documentation effort by 80%.

SELECTED PROJECTS

FinPilot AI | Multi-Agent Reasoning Platform

Multi-agent orchestration with LangGraph ReAct, FastMCP ingestion (80% latency reduction), and DeepEval automated evaluation (93% score).

ALAN-GPT | Enterprise Knowledge & Model Search Engine

High-accuracy vector indexing and semantic retrieval platform covering 500+ ML models (91% retrieval accuracy).

EDUCATION

Bachelor of Science in Statistics and Computer Science – Jagruti Degree and PG College — **Jul 2018 -- Jul 2021**